

Monthly Status Report

E-Pharma Management System — Consolidated Summary (Phases 1–4)

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Project	Techmentee — E-Pharma Management System	Coordinator	Vaishali Pujari
Reporting Period	13 July 2026 – 02 August 2026 (Phases 1–4)	Report Date	7 August 2026
Phases Covered	Weekly reports: Wk 1, Wk 2, Phase 3, Phase 4	Overall Status	On Track — 4 of 6 phases complete

1. Executive Summary

This report consolidates all weekly status reports submitted to date (Week 1, Week 2, Phase 3 and Phase 4). Four of the six planned phases are complete and verified end-to-end, ahead of the original schedule. The E-Pharma Management System is a working, integrated platform connecting patients, doctors, pharmacies and administrators, with real payment processing, event-driven notifications, teleconsultation, and a full administrative reporting and content-management suite. Throughout, the guiding principle has been to build production-shaped integrations behind clean configuration seams: every third-party service (payments, Kafka, SMS/email) runs in a zero-config demo mode and switches to the live provider through configuration alone, so the system is always demonstrable while remaining production-ready.

2. Phase-by-Phase Progress

Phase	Scope	Status	Key deliverables
1	Project understanding & system analysis	Completed	System Analysis & Development Plan; working end-to-end prototype covering all four roles (patient, doctor, pharmacy, admin)
2	Core module hardening	Completed	OTP-verified patient registration; verification-document upload for doctors & pharmacies; input-validation & error-handling pass; coding standards
3	Commerce & consultation	Completed	Stripe/Razorpay payments (signature-verified); Kafka event streaming; teleconsultation chat + Jitsi video; refill reminders
4	Admin ERP & reporting	Completed	Revenue/consultation/order reports; SMS/email gateway on the Kafka stream; category & specialty management; admin-editable CMS pages
5	Performance & security	Planned	PostgreSQL migration; encryption at rest; audit logs; load testing; accessibility pass
6	Deployment & handover	Planned	Production deployment; final documentation; demo & handover

3. Delivery Metrics

Phases completed	4 of 6 (Phases 1–4) — ahead of schedule
Application roles	Patient · Doctor · Pharmacy · Admin (role-based access on every route)
Automated tests	45 end-to-end assertions, all passing (npm test)
Payment providers	Stripe (default) & Razorpay — signature-verified, provider is a config switch
Event streaming	In-process bus by default; Apache Kafka via configuration

Database tables	13 (users, medicines, orders, order_items, appointments, prescriptions, notifications, payments, messages, refill_reminders, taxonomy, cms_pages, otps)
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4. Cumulative Capabilities Delivered

- Authentication & RBAC — token sessions, bcrypt-hashed passwords, OTP-verified patient sign-up, admin approval of doctors & pharmacies with document verification
- Commerce — medicine catalog, cart, stock-safe orders with server-computed totals, signature-verified Stripe/Razorpay payments, order pipeline and refill reminders
- Teleconsultation — appointment booking, party-scoped chat and Jitsi video rooms, doctor e-prescriptions and earnings
- Event-driven backbone — orders, payments and notifications published to topics; notification persistence and SMS/email delivery run as independent consumers
- Administration — KPI dashboard, reporting (revenue, top medicines, consultations), catalog (categories/specialties) and CMS (FAQ/Terms/Privacy) management

5. Key System-Design Decisions (across the month)

- Configuration-driven integrations — payments, Kafka and SMS/email each ship a mock and switch to the live provider via environment variables; the demo never needs external infrastructure
- Event-driven decoupling — producers (orders/payments) are separated from consumers (notifications, receipts, delivery) via topics, so new services attach without changing business logic
- Provider-neutral payments — a single verified-checkout flow spans Stripe and Razorpay; the provider is a config switch, not a code change
- Non-breaking enhancements — managed catalog is advisory (datalist, not a hard constraint) and the database schema is plain SQL, keeping a clean migration path to PostgreSQL
- Testable by construction — a single end-to-end suite (45 assertions) exercises every role and every money/security path against a throwaway database

6. Plan for the Coming Month (Phases 5–6)

- Phase 5 — Performance & security: PostgreSQL migration, encryption at rest for medical records, audit logging, load testing, accessibility pass
- Phase 6 — Deployment & handover: production deployment, final documentation, demo and handover

7. Issues / Blockers

None blocking; the system is fully functional in demo mode. Decisions requested before production hardening: (1) target hosting environment for the PostgreSQL migration and a managed Kafka cluster; (2) the payment provider to license for go-live (Stripe or Razorpay, both integrated); and (3) the SMS/email provider (Twilio or AWS SES) for live notification delivery.

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